



UNSW Course Outline

COMP1521 Computer Systems Fundamentals - 2026

Published on the 01 Jun 2026

General Course Information

Course Code : COMP1521

Year : 2026

Term : Term 2

Teaching Period : T2

Is a multi-term course? : No

Faculty : Faculty of Engineering

Academic Unit : School of Computer Science and Engineering

Delivery Mode : In Person

Delivery Format : Standard

Delivery Location : Kensington

Campus : Sydney

Study Level : Postgraduate, Undergraduate

Units of Credit : 6

Useful Links

[Handbook Class Timetable](#)

Course Details & Outcomes

Course Description

This course introduces students to how computer systems are structured in terms of basic electronic components, how they are used to implement procedural programs, and how they are structured as a collection of software layers. It introduces students to low-level software layers

such as operating systems, and introduces concurrency concepts. The goal is to give students a solid understanding of what happens when high-level programs are executed, as a basis for further study in important areas of computing such as computer architecture, operating systems.

Course Aims

- Provide a programmer's view on how a computer system executes programs, manipulates data and communicates
- Enable students to become effective programmers in dealing with issues of performance, portability, and robustness
- Serve as a foundation for later courses on networks, operating systems, computer architecture and compilers, where a deeper understanding of systems-level issues is required

Relationship to Other Courses

Before commencing this course, students should be able to:

- write simple programs in the C programming language
- define and invoke functions and return results in C
- define and manipulate structured data in C
- use pointers to access data objects

These are assumed to have been acquired in COMP1511 or COMP1911.

Course Learning Outcomes

Course Learning Outcomes
CL01 : Describe the architectural layers (fundamental parts) of a modern computer system from hardware device (chip) levels upwards.
CL02 : Explain how the major components of a CPU work together, including how data and instructions are represented in a computer.
CL03 : Design, implement, and analyse small programs at the assembly/machine level.
CL04 : Describe the relationship between a high-level procedural language (C) and assembly (machine language) which implements it, including how a compiled program is executed in a classical von Neumann machine.
CL05 : Describe the components comprising, and the services offered by, an operating system.
CL06 : Implement simple programs involving communication and concurrency.

Course Learning Outcomes	Assessment Item
CL01 : Describe the architectural layers (fundamental parts) of a modern computer system from hardware device (chip) levels upwards.	• Labs • Tests • Final Exam
CL02 : Explain how the major components of a CPU work together, including how data and instructions are represented in a computer.	• Labs • Tests • Final Exam
CL03 : Design, implement, and analyse small programs at the assembly/machine level.	• Labs • Tests • Assignments 1 and 2 • Final Exam
CL04 : Describe the relationship between a high-level procedural language (C) and assembly (machine language) which implements it, including how a compiled program is executed in a classical von Neumann machine.	• Labs • Tests • Assignments 1 and 2 • Final Exam
CL05 : Describe the components comprising, and the services offered by, an operating system.	• Tests • Final Exam
CL06 : Implement simple programs involving communication and concurrency.	• Tests • Final Exam

Learning and Teaching Technologies

Moodle - Learning Management System | Custom LMS | YouTube | Blackboard Collaborate | Echo 360

Learning and Teaching in this course

This course uses the standard set of practice-focussed teaching strategies employed by most

CSE foundational courses:

- Lectures ... introduce concepts, show examples
- Tutorials ... reinforce concepts and provide additional examples
- Lab Work ... provide examples of using various technologies
- Assignments ... allow you to solve larger problems

Having said that, the second half of the course is more discursive than other CSE foundational courses.

This course is taught the way it is because it aims to give a broad view of many topics in computer systems, to provide a foundation for further study in later systems-related courses. At the same time, it provides further practice in developing software, but at a level closer to the machine than other foundational courses.

Other Professional Outcomes

This course contributes to the development of the following graduate capabilities:

scholarship: understanding of their discipline in its interdisciplinary context	lectures, assignments
scholarship: capable of independent and collaborative enquiry	lab work, assignments
scholarship: rigorous in their analysis, critique, and reflection	tutorials
scholarship: able to apply their knowledge and skills to solving problems	tutorials, lab work, assignments
scholarship: ethical practitioners	all course-work, by doing it yourself
scholarship: capable of effective communication	tutorials
scholarship: digitally literate	everywhere in CSE
leadership: enterprising, innovative and creative	assignments
leadership: collaborative team workers	lab work, assignments
professionalism: capable of operating within an agreed Code of Practice	all prac work

Assessments

Assessment Structure

Assessment Item	Weight	Relevant Dates
Labs Assessment Format: Individual	15%	Start Date: Monday each week, Week 1 - Week 10, Except Week 6 Due Date: 1 Week After Start Date
Tests Assessment Format: Individual	10%	Start Date: Thursday Each Week, Week 3 - Week 10 Due Date: 1 Week After Start Date
Assignments 1 and 2 Assessment Format: Individual	30%	Due Date: Assignment 1 - Week 5, Assignment 2 - Week 10
Final Exam Assessment Format: Individual	45%	Start Date: Final Exam Period Due Date: Final Exam Period

Assessment Details

Labs

Assessment Overview

Following the tutorial class each week, there will be a two-hour laboratory class, during which you will work on a variety of small practical problems involving the tools introduced in lectures. Because this course has a significant practical component, laboratory classes are **important**. If you do not put a good amount of effort into the lab classes, you risk failing the final exam.

Each week, there will be one or more exercises to work on. These exercises will be released in the week preceding the lab class. Each exercise should take 2-3 hours to complete.

During the lab, your tutor will provide feedback on your approach to the problem and on the style of your solution. Some labs may contain exercises which will be assessed during the lab.

Completed exercises need to be submitted. You must submit exercises before the deadline using **give** to obtain a mark for a lab exercise, The usual lab exercise submission deadline will be 12:00 (midday) Monday some lab exercises may have an extended deadline.

The lab exercises for each week are worth in total 2 marks. All of your lab marks will be summed to give you a mark out of 18; if their sum exceeds 15, your total mark will be capped at 15.

Most labs include one or more challenge exercises. Challenge exercises may involve concepts not covered in lectures and they range in difficulty from not-very-hard to almost-impossible.

The contribution of challenge exercises to lab marks will be limited to 20%; hence you can obtain nearly all (over 95%) marks available for the lab component without completing challenge exercises.

If you wish to obtain a high mark for COMP1521, attempting some challenge exercises is highly recommended.

If your goal is just to master the core material and pass COMP1521, you can ignore challenge exercises.

Course Learning Outcomes

- CLO1 : Describe the architectural layers (fundamental parts) of a modern computer system from hardware device (chip) levels upwards.
- CLO2 : Explain how the major components of a CPU work together, including how data and instructions are represented in a computer.
- CLO3 : Design, implement, and analyse small programs at the assembly/machine level.
- CLO4 : Describe the relationship between a high-level procedural language (C) and assembly (machine language) which implements it, including how a compiled program is executed in a classical von Neumann machine.

Submission notes

Submission via CSE give

Assignment submission Turnitin type

This is not a Turnitin assignment

Generative AI Permission Level

No Assistance

This assessment is designed for you to complete without the use of any generative AI. You are not permitted to use any generative AI tools, software or service to search for or generate information or answers.

For more information on Generative AI and permitted use please see [here](#).

Tests

Assessment Overview

There will be weekly tests from weeks 3-10. designed to give you timely and realistic feedback of your understanding of the course material. Tests may be programming exercises, multiple choice questions, or both. Each test should take around 1 hour.

These will be conducted in your own time under self-enforced exam-like conditions. Each test will specify the conditions, but typically these will include:

1. no assistance permitted from any person;
2. a time limit;
3. no access to materials (written or online) except specified language documentation or man pages.

Each test is worth 1.7 marks, and will be automarked. Your total mark for the tests components is computed as a sum of your best 6 of 8 test marks.

Course Learning Outcomes

- CLO1 : Describe the architectural layers (fundamental parts) of a modern computer system from hardware device (chip) levels upwards.
- CLO2 : Explain how the major components of a CPU work together, including how data and instructions are represented in a computer.
- CLO3 : Design, implement, and analyse small programs at the assembly/machine level.
- CLO4 : Describe the relationship between a high-level procedural language (C) and assembly (machine language) which implements it, including how a compiled program is executed in a classical von Neumann machine.
- CLO5 : Describe the components comprising, and the services offered by, an operating system.
- CLO6 : Implement simple programs involving communication and concurrency.

Assessment Length

1 hour

Submission notes

Submission via CSE give

Assignment submission Turnitin type

This is not a Turnitin assignment

Generative AI Permission Level

No Assistance

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Assignments 1 and 2

Assessment Overview

There are two assessable programming assignments. Assignments give you the chance to practice what you have learnt on relatively large problems (compared to the small exercises in the labs). Assignments are a *very important* part of this course, therefore it is essential that you attempt them yourself.

- Assignment 1, on Assembly programming (MIPS) worth 15%
- Assignment 2, on System programming worth 15%

Each assignment should take 10-15 hours to complete.

Assignments are primarily automarked, with feedback from tutors on programming style.

Course Learning Outcomes

- CLO3 : Design, implement, and analyse small programs at the assembly/machine level.
- CLO4 : Describe the relationship between a high-level procedural language (C) and assembly (machine language) which implements it, including how a compiled program is executed in a classical von Neumann machine.

Detailed Assessment Description

- Assignment 1, on Assembly programming (MIPS), worth 15% - Due End of Week 5
- Assignment 2, on System programming, worth 15% - Due Week 10

Submission notes

Submission via CSE give

Assignment submission Turnitin type

This is not a Turnitin assignment

Generative AI Permission Level

No Assistance

This assessment is designed for you to complete without the use of any generative AI. You are not permitted to use any generative AI tools, software or service to search for or generate information or answers.

For more information on Generative AI and permitted use please see [here](#).

Final Exam

Assessment Overview

There will be a three-hour final exam, held in the CSE labs during the exam period. The exam runs in a closed environment (no internet) but with relevant documentation provided.

During this exam you will need to execute, debug and test your answers to implementation tasks which will be similar to those encountered in lab exercises and weekly tests.

There is a hurdle requirement on the final exam. If you do not score at least 40% (18.0/45) on the exam (after scaling), you cannot pass this course. If your overall course score exceeds 50%, despite scoring very poorly (<40%) on the exam, the hurdle will be enforced via a grade of UF.

Course Learning Outcomes

- CLO1 : Describe the architectural layers (fundamental parts) of a modern computer system from hardware device (chip) levels upwards.
- CLO2 : Explain how the major components of a CPU work together, including how data and instructions are represented in a computer.
- CLO3 : Design, implement, and analyse small programs at the assembly/machine level.
- CLO4 : Describe the relationship between a high-level procedural language (C) and assembly (machine language) which implements it, including how a compiled program is executed in a classical von Neumann machine.
- CLO5 : Describe the components comprising, and the services offered by, an operating system.
- CLO6 : Implement simple programs involving communication and concurrency.

Assessment Length

3 hours

Assignment submission Turnitin type

This is not a Turnitin assignment

Hurdle rules

There is a hurdle requirement on the final exam. If you do not score at least 40% (18.0/45) on the exam (after scaling), you cannot pass this course. If your overall course score exceeds 50%, despite scoring very poorly (<40%) on the exam, the hurdle will be enforced via a grade of UF.

Generative AI Permission Level

No Assistance

This assessment is designed for you to complete without the use of any generative AI. You are

not permitted to use any generative AI tools, software or service to search for or generate information or answers.

For more information on Generative AI and permitted use please see [here](#).

General Assessment Information

Laboratory Classes

Following the tutorial class each week, there will be a two-hour laboratory class, during which you will work on a variety of small practical problems involving the tools introduced in lectures.

Because this course has a significant practical component, laboratory classes are **important**. If you do not put a good amount of effort into the lab classes, you risk failing the final exam.

Each week, there will be one or more exercises to work on. These exercises will be released in the week preceding the lab class. During the lab, your tutor will provide feedback on your approach to the problem and on the style of your solution. Some labs may contain exercises which will be assessed during the lab.

In addition, **students must complete a specific check-off exercise during their scheduled lab session**; this task must be verified by a tutor in person (including via a live online Teams meeting for online classes) for the remaining take-home exercises to contribute toward the weekly laboratory mark. While these exercises are not conducted under exam conditions and lab assistants will be available to provide guidance, they are a mandatory requirement designed to ensure you are consistently engaging with the core concepts in preparation for the final exam.

Completed exercises need to be submitted. You must submit exercises before the deadline using **give** to obtain a mark for a lab exercise. The usual lab exercise submission deadline will be 12:00 (midday) Monday. Some lab exercises may have an extended deadline.

The lab exercises for each week are worth in total 2 marks. All of your lab marks will be summed to give you a mark out of 18; if their sum exceeds 15, your total mark will be capped at 15.

Most labs include one or more challenge exercises. Challenge exercises may involve concepts not covered in lectures and they range in difficulty from not-very-hard to almost-impossible.

The contribution of challenge exercises to lab marks will be limited to 20%; hence you can obtain nearly all (over 95%) marks available for the lab component without completing challenge exercises.

If you wish to obtain a high mark for COMP1521, attempting some challenge exercises is highly recommended.

If your goal is just to master the core material and pass COMP1521, you can ignore challenge exercises.

Late lab submissions will be penalised. The exact penalty will be the standard UNSW late penalty of a mark reduction equal to 5% of the achieved (raw) assessment mark per day late, or a zero mark after 5 days (i.e. 120 hours).

Assignments

There are two assessable programming assignments. Assignments give you the chance to practice what you have learnt on relatively large problems (compared to the small exercises in the labs). Assignments are a *very important* part of this course, therefore it is essential that you attempt them yourself.

- Assignment 1, on Assembly programming (MIPS); due Friday Week 5; worth 15%
- Assignment 2, on System programming; due Friday Week 9; worth 15%

Late assignment submissions will be penalised. The exact penalty will be the standard UNSW late penalty of a mark reduction equal to 5% of the maximum assessment mark per day late, or a zero mark after 5 days (i.e. 120 hours).

Weekly Tests

There will be weekly tests from weeks 3–10 designed to give you timely and realistic feedback of your understanding of the course material. Tests may be programming exercises, multiple choice questions, or both.

These will be conducted in your own time under self-enforced exam-like conditions. Each test will specify the conditions, but typically these will include:

1. no assistance permitted from any person;
2. a time limit;
3. no access to materials (written or online) except specified language documentation or man pages.

Each test is worth 1.7 marks, and will be automarked. Your total mark for the tests component is computed as a sum of your best 6 of 8 test marks.

Late lab submissions will be penalised. The exact penalty will be the standard UNSW late penalty of a mark reduction equal to 5% of the achieved (raw) assessment mark per day late, or a zero mark after 5 days (i.e. 120 hours).

Final Exam

There will be a three-hour final exam, held in the CSE labs during the exam period. This will be centrally timetabled, and appear in your UNSW exam timetable.

During this exam you will need to execute, debug and test your answers to implementation tasks which will be similar to those encountered in lab exercises and weekly tests.

There is a hurdle requirement on the final exam. If you do not score at least 40% (18.0/45) on the exam (after scaling), you cannot pass this course. If your overall course score exceeds 50%, despite scoring very poorly (<40%) on the exam, the hurdle will be enforced via a grade of UF.

Grading Basis

Standard

Requirements to pass course

There is a hurdle requirement on the final exam. If you do not score at least 40% (18.0/45) on the exam (after scaling), you cannot pass this course. If your overall course score exceeds 50%, despite scoring very poorly (<40%) on the exam, the hurdle will be enforced via a grade of UF.

Course Schedule

Teaching Week/Module	Activity Type	Content
Week 1 : 1 June - 7 June	Lecture	Course intro; MIPS assembly programming
	Tut-Lab	C revision
Week 2 : 8 June - 14 June	Lecture	MIPS assembly programming
	Tut-Lab	MIPS assembly programming
Week 3 : 15 June - 21 June	Lecture	MIPS assembly programming
	Tut-Lab	MIPS assembly programming
Week 4 : 22 June - 28 June	Lecture	Integer representations; Bit manipulations
	Tut-Lab	MIPS assembly programming
Week 5 : 29 June - 5 July	Lecture	Bit manipulations; IEEE-754 floating point representation
	Tut-Lab	Integer representations; Bit manipulations
Week 6 : 6 July - 12 July	Other	flexibility week
Week 7 : 13 July - 19 July	Lecture	File systems
	Tut-Lab	Bit manipulations; Floating point representation
Week 8 : 20 July - 26 July	Lecture	File systems; File metadata; UTF8 character encoding
	Tut-Lab	Manipulating files and file metadata
Week 9 : 27 July - 2 August	Lecture	Concurrency and parallelism using processes and threads
	Tut-Lab	Manipulating files, metadata, directories, UTF8
Week 10 : 3 August - 9 August	Lecture	Final exam
	Tut-Lab	Working with processes in C; using threads in C

Attendance Requirements

Students are strongly encouraged to attend all classes and review lecture recordings.

Course Resources

Recommended Resources

There is no single book that covers all of the material in this course at the right level of detail and using the same technology base as we are. The lecture notes should provide sufficient detail to introduce topics, and you will then study them in further depth in the tutes, labs and assignments.

There are also many online resources available, and we will provide links to the most useful ones. Some are listed below. If you find others, please post links in the Comments section on the Course Outline page.

The following is a Recommended Reading for this course:

- *Computer Systems: A Programmer's Perspective*, by Randal E. Bryant and David R. O'Hallaron; Prentice-Hall ([web site](#))

There are copies in the UNSW Bookstore and in the library. It covers many of the topics in the

course, but uses a different machine architecture (i.e., not MIPS).

Some suggestions for other books that cover at least some of the topics in this course:

- *Introduction to Computer Systems: From Bits and Gates to C and Beyond*, by Yale N. Patt and Sanjay J. Patel; McGraw Hill
- *The Elements of Computing Systems: Building a Modern Computer from First Principles*, by Noam Nisan and Shimon Schocken; MIT Press ([web site](#), including lecture slides)

Documentation for the various systems used in the course is linked from the course website.

Course Evaluation and Development

Every term, student feedback is requested in a survey using UNSW's myExperience online survey system where the feedback will be used to make improvements to the course. This is used to assess the quality of the course so that we can make ongoing improvements. We take this feedback seriously and use it to improve the course materials and their delivery. Students are also encouraged to provide informal feedback during the term by letting the Lecturer in Charge or any of the course staff know of any problems, as soon as they arise. Suggestions will be listened to openly, positively, constructively, and thankfully, and every reasonable effort will be made to address them.

Recent myExperience evaluations showed that students were highly satisfied with most aspects of the course. However, there are always things that can be improved and this term we are fine tuning and improving tutorial, lab and revision material to provide more support for the later more complex topics.

CSE may also run its own survey, midway through the term, to elicit feedback while courses are still running. This course improves only because we see the difficulties that students experience and attempt to adjust things so that you get to learn what/how you need. If anything's not working for you, please let us know and we'll do whatever we can to help and hopefully help students in later cohorts as well.

Staff Details

Position	Name	Email	Location	Phone	Availability	Equitable Learning Services Contact	Primary Contact
Administrator	COURSE EMAIL					Yes	Yes
Lecturer	Tara Hamilton					No	No
Convenor	Angela Finlayson					No	No
Administrator	Alex Blackmore					No	No
	Anna Brew					No	No
	Sophie Fox					No	No

Other Useful Information

Academic Information

I. Special consideration and supplementary assessment

If you have experienced an illness or misadventure beyond your control that will interfere with your assessment performance, you are eligible to apply for Special Consideration prior to, or within 3 working days of, submitting an assessment or sitting an exam.

Please note that UNSW has a Fit to Sit rule, which means that if you sit an exam, you are declaring yourself fit enough to do so and cannot later apply for Special Consideration.

For details of applying for Special Consideration and conditions for the award of supplementary assessment, please see the information on UNSW's [Special Consideration page](#).

II. Administrative matters and links

All students are expected to read and be familiar with UNSW guidelines and policies. In particular, students should be familiar with the following:

- [Attendance](#)
- [UNSW Email Address](#)
- [Special Consideration](#)
- [Exams](#)
- [Approved Calculators](#)
- [Academic Honesty and Plagiarism](#)
- [Equitable Learning Services](#)

III. Equity and diversity

Those students who have a disability that requires some adjustment in their teaching or learning environment are encouraged to discuss their study needs with the course convener prior to, or at the commencement of, their course, or with the Equity Officer (Disability) in the Equitable Learning Services. Issues to be discussed may include access to materials, signers or note-takers, the provision of services and additional exam and assessment arrangements. Early notification is essential to enable any necessary adjustments to be made.

IV. Professional Outcomes and Program Design

Students are able to review the relevant professional outcomes and program designs for their streams by going to the following link: <https://www.unsw.edu.au/engineering/student-life/student-resources/program-design>.

Note: This course outline sets out the description of classes at the date the Course Outline is published. The nature of classes may change during the Term after the Course Outline is published. Moodle or your primary learning management system (LMS) should be consulted for the up-to-date class descriptions. If there is any inconsistency in the description of activities between the University timetable and the Course Outline/Moodle/LMS, the description in the Course Outline/Moodle/LMS applies.

Academic Honesty and Plagiarism

UNSW has an ongoing commitment to fostering a culture of learning informed by academic integrity. All UNSW students have a responsibility to adhere to this principle of academic integrity. Plagiarism undermines academic integrity and is not tolerated at UNSW. *Plagiarism at UNSW is defined as using the words or ideas of others and passing them off as your own.*

Plagiarism is a type of intellectual theft. It can take many forms, from deliberate cheating to accidentally copying from a source without acknowledgement. UNSW has produced a website with a wealth of resources to support students to understand and avoid plagiarism, visit: student.unsw.edu.au/plagiarism. The Learning Centre assists students with understanding academic integrity and how not to plagiarise. They also hold workshops and can help students one-on-one.

You are also reminded that careful time management is an important part of study and one of the identified causes of plagiarism is poor time management. Students should allow sufficient time for research, drafting and the proper referencing of sources in preparing all assessment tasks.

Repeated plagiarism (even in first year), plagiarism after first year, or serious instances, may also be investigated under the Student Misconduct Procedures. The penalties under the procedures can include a reduction in marks, failing a course or for the most serious matters (like plagiarism in an honours thesis or contract cheating) even suspension from the university. The Student Misconduct Procedures are available here:

www.gs.unsw.edu.au/policy/documents/studentmisconductprocedures.pdf

Submission of Assessment Tasks

Work submitted late without an approved extension by the course coordinator or delegated authority is subject to a late penalty of five percent (5%) for that assessment item.

The late penalty is applied per calendar day (including weekends and public holidays) that the assessment is overdue. There is no pro-rata of the late penalty for submissions made part way through a day. This is for all assessments where a penalty applies.

Work submitted after five days (120 hours) will not be accepted and a mark of zero will be awarded for that assessment item.

For some assessment items, a late penalty may not be appropriate. These will be clearly indicated in the course outline, and such assessments will receive a mark of zero if not completed by the specified date. Examples include:

- Weekly online tests or laboratory work worth a small proportion of the subject mark;
- Exams, peer feedback and team evaluation surveys;
- Online quizzes where answers are released to students on completion;
- Professional assessment tasks, where the intention is to create an authentic assessment that has an absolute submission date; and,
- Pass/Fail assessment tasks.

Faculty-specific Information

[Engineering Student Support Services](#) – The Nucleus - enrolment, progression checks, clash requests, course issues or program-related queries

[Engineering Industrial Training](#) – Industrial training questions

[UNSW Study Abroad](#) – study abroad student enquiries (for inbound students)

[UNSW Exchange](#) – student exchange enquiries (for inbound students)

[UNSW Future Students](#) – potential student enquiries e.g. admissions, fees, programs, credit transfer

Phone

(+61 2) 9385 8500 – Nucleus Student Hub

School Contact Information

CSE Help! - on the Ground Floor of K17

- For assistance with coursework assessments.

The Nucleus Student Hub - <https://nucleus.unsw.edu.au/en/contact-us>

- Course enrolment queries.

Grievance Officer - grievance-officer@cse.unsw.edu.au

- If the course convenor gives an inadequate response to a query or when the courses convenor does not respond to a query about assessment.

Student Reps - stureps@cse.unsw.edu.au

- If some aspect of a course needs urgent improvement. (e.g. Nobody responding to forum queries, cannot understand the lecturer)

You should **never** contact any of the following people directly:

- Vice Chancellor
- Pro-vice Chancellor Education (PVCE)
- Head of School
- CSE administrative staff
- CSE teaching support staff

They will simply bounce the email to one of the above, thereby creating an unnecessary level of

indirection and a delay in the response.